

Practical - 11

Aim: Given string num representing a non- negative integer num, and an integer k, return the smallest possible integer after removing k digits from num.

PROGRAM CODE :

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
char* removeKdigits(char* num, int k) {
    int len = strlen(num);
    int stack_size = 0;
    char* stack = (char*)malloc(len + 1);
    if (stack == NULL) {
        printf("Memory allocation failed.\n");
        exit(1);
    }
    for (int i = 0; i < len; i++) {
        while (stack_size > 0 && k > 0 && stack[stack_size - 1] > num[i]) {
            stack_size--;
            k--;
        }
        stack[stack_size++] = num[i];
    }
    stack_size -= k;
    stack[stack_size] = '\0';
    int start = 0;
    while (stack[start] == '0' && start < stack_size) {
        start++;
    }
}
```

```
if (start == stack_size) {  
    strcpy(stack, "0");  
} else {  
    memmove(stack, stack + start, stack_size - start + 1);  
}  
return stack;  
}  
int main() {  
    char *num = NULL;  
    int k;  
    size_t buffer_size = 0;  
    printf("Enter the number: ");  
    getline(&num, &buffer_size, stdin);  
    num[strcspn(num, "\n")] = '\0';  
    printf("Enter the value of k: ");  
    scanf("%d", &k);  
    char* result = removeKdigits(num, k);  
    printf("Result: %s\n", result);  
    free(result);  
    free(num);  
    return 0;  
}
```

OUTPUT :-

Enter the number: 123456789

Enter the value of k: 8

Result: 1

main.c	Output
<pre>21 int start = 0; 22 while (stack[start] == '0' && start < stack_size) { 23 start++; 24 } 25 if (start == stack_size) { 26 strcpy(stack, "0"); 27 } else { 28 memmove(stack, stack + start, stack_size - start + 1); 29 } 30 return stack; 31 } 32 int main() { 33 char *num = NULL; 34 int k; 35 size_t buffer_size = 0; 36 printf("Enter the number: "); 37 getline(&num, &buffer_size, stdin); 38 num[strcspn(num, "\n")] = '\0'; 39 printf("Enter the value of k: "); 40 scanf("%d", &k); 41 char* result = removeKdigits(num, k); 42 printf("Result: %s\n", result); 43 free(result); 44 }</pre>	<pre>/tmp/I1kUv1UppD.o Enter the number: 123456789 Enter the value of k: 8 Result: 1 === Code Execution Successful ===</pre>